



This is me

05.12.2000

I am an enthusiastic electrical engineering student with a strong attention to detail and a great desire to understand the depths of my field.

My academic journey has provided me with the technical knowledge to approach new topics in a goal-oriented manner through various methods. But more importantly, it showed my dedication and capability of quickly adapting to new concepts and technologies.

Contact



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Leon Nachtrab



leon-nachtrab.com

Languages

German	Mother tongue
English	C1
French	B2

Coding

C++/C
Python
MATLAB

Leon Nachtrab

Curriculum Vitae

Experience

2026
-
present

Vision-Based Automation Engineer Intern

Swiss Engineering Partners AG, Zurich Switzerland

Lead computer vision and automation for pilot project for World Travel Catering & Onboard Services Expo 2026

Education

2023
-
2025

Master's Degree in Electrical Engineering and Information Technology

ETH Zürich

Focus	Embedded Systems Electric Machines Control Machine Learning and Computer Vision Power Electronics
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Master's Thesis	Embedded Control System for Bio-Inspired Saccadic Camera Motion
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Overall Grade	5.7
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2020
-
2024

Bachelor's Degree in Electrical Engineering and Information Technology

ETH Zürich

Core Subjects	Control Systems Power Electronics Measurement and Testing Technology
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Bachelor's Thesis	Design and Commissioning of a Double Bearingless Outrunner Motor for Fluid Manipulation Winner of Swissmem Best Thesis Award 2023 – Industry sector drive technology
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Overall Grade	5.44
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2013
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2019

High School Diploma (Matura)

Cantonal School of Obwalden

Major	Physics and Applied Mathematics
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Software

STMicroelectronics Tools

Altium Designer

SolidWorks

Autodesk Fusion 360

MATLAB/Simulink

Plexim

GitHub

Skillset

Embedded Systems and Sensor
Networks (C/C++, STM32, FreeRTOS)

PCB Design & Hardware Integration

Quick Prototyping & Iterative
Development

Training Machine Learning Models

3D Modelling

Project Leadership & Ownership

Client-Facing Communication &
Consulting

On a personal Note

"To handle yourself, use your head.
To handle others, use your heart."
— Eleanor Roosevelt

"To handle everything else, use
PID control."

What ignites my interest and lets
me forget the time is a problem
that requires going back and forth
between the theoretical software
and the physical hardware. This
interaction is where my true
professional interest lies.

Outside of engineering, you'll find
me ski touring in the Alps,
mountain biking through trails,
playing badminton, making music,
or experimenting in the kitchen.

Ventures

- 2025 ● **Master's Thesis – Collaboration Arma Suisse**
*Design of a Custom Dual Embedded Motor Driver for PMSM in
Military UAV Detection Applications*
- 2024 ● **Exploration Lab – Collaboration with Bühler Group,
Bossard AG, VZug AG and VAT Group**
- Numerous smaller Projects within a Consulting Scope
- 2025 ● **AIWorkflowGenerator – Project at Bossard AG**
*AI Driven PDF Conversion Automating Instruction Set Creation for
Bossard Smart Stations*
- 2024 ● **Battery Retrofit – Project at Bossard AG**
*Engineering Solution for the SmartBins Reducing Costs by 50% and
CO₂ by 16% - Now in Industrialization Phase*
- 2023 ● **Bachelor's Thesis – Collaboration with Levitronix**
Bearingless Double Outrunner Control
- 2021 ● **Tutor in Digital Circuits**
ETH Zürich
- 2019 ● **Military Service**
- *Infantry DD Ber Bat 104*
2020 *COVID-19 Deployment in Geneva*
- 2020 ● **ASVZ TL – Badminton Coach**
- *Coaching and organizing group badminton sessions at ASVZ.*
present

References

- **Philipp Mayer**
Postdoctoral Researcher at ETH Zürich
Supervisor of the Master's Thesis
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- **Julian Moosmann**
PhD Candidate at ETH Zürich
Supervisor of the Master's Thesis
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- **Daniel Giesler**
EXL LEAD at ETH Zürich and Innovation Manager at Bossard AG
Supervisor for Projects at Bossard AG
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- **Severin Hiltbrunner**
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